

Exploring the Use of the SoundVizVR Plugin with Game Developers in the Development of Sound-Accessible Virtual Reality Games

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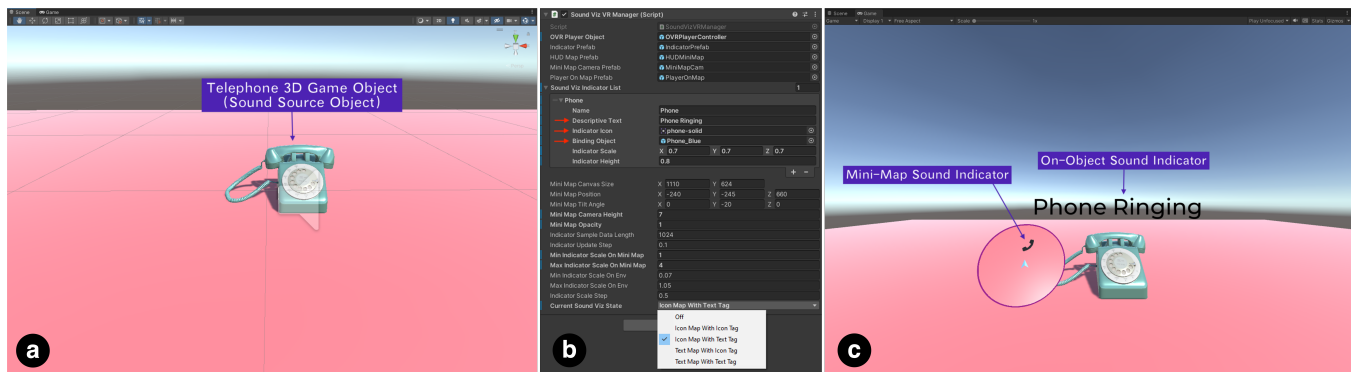


Figure 1: The SoundVizVR Plugin is integrated into a Unity VR project. (a) A Unity Editor View shows a 3D telephone game object as a sound source object; (b) The SoundVizVR Plugin config is provided in the “SoundVizVR Manager” prefab, which is the manager of the plugin. The red arrows highlight some of the critical parameters for a sound indicator setup; (c) A Unity in-game preview of the SoundVizVR Plugin with icon sound indicators on the Head-Up Display (HUD) Mini-Map and text sound indicators above the sounding object.

ABSTRACT

Based on a virtual reality (VR) sound visualization indicator system named “SoundVizVR” proposed and evaluated in a previous research, we developed a “SoundVizVR Plugin” for the Unity platform which enabled developers to adopt the SoundVizVR system in their VR applications and games. In this study, we worked with two VR game developers to explore the usability of the SoundVizVR Plugin and how it impacted the view on accessibility from the game developers’ perspectives. The developers used the SoundVizVR Plugin for one week to integrate it into their current projects or design new games with the plugin. According to the findings of the study process, we provide game design practices for making sounds in VR games accessible and suggestions on optimizing the developer experience of a sound accessibility plugin for the Unity platform.

CCS CONCEPTS

• **Human-centered computing** → **User studies**; *Accessibility systems and tools*.

KEYWORDS

virtual reality, game development, accessibility, development tool, audio visualization, deaf and hard-of-hearing, sound accessibility

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1 INTRODUCTION

Sounds are used in virtual reality (VR) environments to provide notifications and interaction feedback [10]. The information conveyed by sounds is critical in providing interaction cues and delivering an immersive experience to VR users. However, acoustic information in VR environment may not be accessible to deaf or hard-of-hearing (DHH) individuals [11, 17]. In this case, sound could be a limiting factor for DHH users to experience the VR content fully.

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Previous research has explored haptic-based [8, 22, 23, 29] and visual-based methods [9, 12, 16] for making everyday sounds accessible for DHH users. Among them, a recent study proposed a new visualization-based method named “SoundVizVR” to make *diegetic* sounds in VR more accessible for DHH users [13]. For conveying sound information of spatial objects in VR environments, the SoundVizVR technique used a combination of *On-Object* and *Mini-Map* sound indicators to visualize sound characteristics, and icons and text to represent sound types (Figure 1.c). While SoundVizVR and other previous methods had presented ways to make VR sounds accessible for DHH users, their usability from a VR game developers’ perspective had not been explored, which can be a critical limitation as it may impact on the adoption of such tools by future VR games and applications. Therefore, based on the proposed SoundVizVR system and the findings from the previous work, we developed a “SoundVizVR Plugin”¹ for the Unity platform² that enables developers to adopt the SoundVizVR sound indicator system in their VR applications and games. With this plugin, developers can promptly visualize different types of sounds that originate from the objects within the VR environments. This “SoundVizVR Plugin” for the Unity platform can assist VR content creators in overcoming the barriers they may face when making the audio of their games and applications accessible for DHH VR users.

In this study, we worked with two VR game developers to evaluate the “SoundVizVR Plugin”. Our primary research question in this study was “*What is the usability of the SoundVizVR Plugin, and how does that impact the accessibility perspectives of VR game developers in their work?*” To address this question, we asked the developers to use this plugin for one week to integrate it into their current projects or new work as they see fit. As a result, we collected insights regarding the content creators’ development experience and design practices when using this plugin. This collected feedback enabled us to explore the usability of the SoundVizVR Plugin and how it impacted the view on accessibility from the developers’ perspectives. In addition, based on those findings, we provided insights on integrating the plugin into VR games to improve audio accessibility.

2 BACKGROUND

2.1 Sound Visualization

Prior work has mainly explored visual and haptic methods to make sounds more accessible for DHH individuals. Here, we focus on prior work closely related to our work that applied visual-based sound accessibility methods.

Sound visualization techniques used in music can convey music information (e.g., pitch, tempo) in real time with animated images [27]. Music visualization methods are widely supported by media player software, such as Windows Media Player, Foobar2000, and iTunes. They visualize music characteristics using spectrum-like or waveform-like 2D displays. There is also research on music visualization techniques conducted among DHH communities to evaluate DHH people’s experience with these methods. For example, S. Nanayakkara et al. implemented a system combining a vibrated chair with a visual display to provide an enhanced musical

experience to DHH people [22]. In addition, J. Mori and D. I. Fels conducted research with DHH people to investigate their emotional reaction to a song with different animated lyrics [21], which indicated that the animated text could provide entertainment value to the music without losing the readability of lyrics.

Similarly, sounds also have vital contributions to the experience designs in games. There are video games that look into using sound visualization methods to make their content more accessible for DHH people. For example, a sandbox video game called “Minecraft” [20] features “Subtitles” that use text labels with arrows to indicate the sound types and the directions of in-game sounds (e.g., rain falls, zombie groans) near the player avatar [30]. Moreover, a battle royale game called “Fortnite” [6] uses a radar-like interface to assist DHH players in accurately locating the vital sound effects in the game environment, like footsteps and gunfire. However, with caption supported [15], the playing experience is still significantly reduced in many commercial video games once sounds are disabled. This limitation may be due to the lacking sound accessibility designs in their major game events (e.g., notifications of shootings from an enemy in a first-person shooter [FPS] game) [4, 32].

2.2 VR Sound Accessibility

In terms of sound accessibility in VR, previous research explores providing accessible acoustic clues using subtitles [1, 7, 25, 28]. Several other research examines VR in the context of storytelling [5] for DHH individuals and learning sign language for DHH children [24]. Previous studies also investigated locating spatial sound sources in VR for DHH users through haptics [11, 17], visuals [11, 13, 19], and multi-modal representation methods (e.g., visual+haptic [18], audio+visual+haptic [19]). Furthermore, some commercial VR games adopt sound accessibility interfaces other than captions, for example, the enemy sound direction indicator in “The Persistence” [14].

Our study is based on prior research from Z. Li et al. that presented a VR sound visualization system for DHH users called “SoundVizVR” [13]. The SoundVizVR system can visualize sound characteristics (e.g., duration, loudness, location) in VR environments via *On-Object* and *Mini-Map* sound indicators. Meanwhile, it can visualize sound types (e.g., phone ringing, footsteps) of sound sources via text and icon representation methods. In this prior study, researchers conducted two empirical evaluations with two groups of DHH participants respectively, proving that the SoundVizVR sound indicator system can assist the DHH users in locating the spatial sound sources, and identifying sound characteristics and sound types of the sound sources in VR scenes.

However, while prior work focused on collecting feedback on the end products with accessibility features integrated, there is little work on evaluating the challenges of integrating an accessibility plugin, especially a sound visualization tool, into VR games during game development. Thus, the exploration of these aspects from game designers’ and developers’ perspectives is demanded.

3 METHODOLOGY

To address our main research question, we conducted a preliminary study with two VR game developers where the developers used the SoundVizVR Plugin over a one-week period and discussed their

¹<https://github.com/ZimingLi/SoundVizVR-Plugin>

²Unity is a development platform for creating interactive 2D and 3D projects.

perspectives on the usability of the plugin and its impact on their accessibility perspectives.

3.1 SoundVizVR Plugin

We implemented the SoundVizVR Plugin package for the Unity platform version 2021.3.3f1 and Oculus Integration version 39.0. The current plugin supports VR projects developed for Meta (Oculus) VR headsets with the Unity platform.

The SoundVizVR Plugin uses icons and text as sound visualization representation methods on a Head-Up Display (HUD) Mini-Map and On-Object indicators to improve the accessibility of diegetic sounds in VR (Figure 1). The plugin enables designers and developers to bind a sound visualization indicator with customizable descriptive text or icons to any sound source object in a Unity scene. Its sound indicator can visualize the sound characteristics such as the existence, loudness, and duration of the sounds originating from a binding game object. With its text and icon representation methods, the sound indicator can additionally convey sound-type information (e.g., gunshots, phone ringing) of the binding object to VR users.

3.1.1 Importing the SoundVizVR Plugin to Unity. We provided a detailed list of instructions for using the SoundVizVR Plugin on the plugin's download page. First, VR designers and developers can import the SoundVizVR Plugin into their existing Unity VR projects as a package³. Then, they can find the manager prefab⁴ of the plugin, named "SoundVizVR Manager", in the package directory and import it into the Unity project scene. Next, designers and developers can configure the plugin by adjusting the parameters in the "SoundVizVR Manager" (Figure 1.b) prefab or programming via the provided Application Program Interface (API) for a more flexible configuration.

3.1.2 Configuring the SoundVizVR Plugin. When a sound source is added to a game object in the VR scene (e.g., a phone ringing sound on a 3D telephone game object, as shown in Figure 1.a), the developer can add it to the "SoundVizVR Manager" by dragging the sound source object to the manager's 'Binding Object' field (Figure 1.b). Then, a *sound indicator* object will be assigned to the sound source object. The *sound indicator* will visually indicate the presence of a sound by showing up in the environment (above the game object of the sound source - in Figure 1.c, an indicator is showing above the telephone) and on the HUD Mini-Map when the object starts making sounds at runtime. The indicator dynamically changes its size to reveal the loudness of the sounds. And the location of the sound can be identified through the Mini-Map and/or the sound indicator above the sound source object. By default, a sound indicator will display as a dynamic red object on top of the sounding game object and the HUD Mini-Map.

To provide descriptive sound information, the developer must define the 'Descriptive Text' and add a representative 'Indicator Icon' to the sound indicator (Figure 1.b). The icons and text can be shown as sound indicators on the HUD Mini-Map (i.e., "Map") and above a sounding game object (i.e., "Tag") based on the developer's preference on the 'Current Sound Viz State' parameter's config

(Figure 1.b). In the example provided in Figure 1.c, the Indicator-Icon of the telephone is shown on the Mini-Map, whereas the Descriptive-Text Tag is shown above the game object.

In addition, the plugin manager object provides flexible options that allow the designers and developers to customize the sound indicators based on their preferences. Regarding the HUD Mini-Map, the plugin enables designers to change the Mini-Map's field of view, size, and opacity, as well as its position and tilt angle on the HUD. As for the sound indicator, the developers can determine the indicator's update rate, the rate of change during its animated transition, and its range of scale on either the HUD Mini-Map or above a sound source. Furthermore, the plugin provides customizability for each sound indicator of a specific binding object. Here, the developers can customize the characteristics of an individual sound indicator, such as its descriptive text content, its icon pattern, its initial scale, and its offset height hovering above an object in the virtual environment.

3.2 Setting and Participants

To explore the usability of the SoundVizVR Plugin and developers' attitudes toward integrating sound accessibility features in their VR games, we conducted a preliminary study in a remote format. It was a week-long study that provided sufficient time for the participants to get familiar with the SoundVizVR Plugin and identify issues and challenges when integrating accessibility features in their games. Two developer participants (two males, aged 24-32) from the authors' institution were chosen for the study.

D1 (Participant 1) had been working as a game developer for about 7 years. He had previously been involved in the development of about 10 digital games. He learned game development skills through self-learning and graduate school. D1 participated in this study because he was interested in game accessibility and believed letting diverse player groups access and enjoy the game experience was vital.

D2 (Participant 2) had been working as a game developer for 11 years. Previously, he had worked on about 8 digital games. He learned game development skills through his university education. D2 participated in this study because he was curious about how to consider, introduce, and standardize accessibility in game design and development.

Furthermore, both participants had experience in VR game development with the Unity platform.

3.3 Study Procedure

The study was approved by the ethics review board of the authors' institution. The study contained a pre-study questionnaire, a week-long study process, a post-study questionnaire, and a follow-up interview. After giving an introduction and having the participant sign the informed consent form, we asked the participant to complete the pre-study questionnaire, which inquired about participants' attitudes and experience in accessibility design during their previous game design process. It took approximately 15 minutes to complete. Next, we sent the SoundVizVR Plugin Unity package and detailed digital user documentation to the participant before the development session. The participant was required to 1) read

³Unity Packages

⁴Unity Prefabs

through the user documentation and get familiar with the SoundVizVR Plugin in the Unity platform; 2) integrate the plugin into their VR applications; 3) configure the plugin (e.g., add icons or text) for the diegetic sound effects in their applications; and, 4) test the VR applications with the visual cues built-in on a VR headset. (Here, an Oculus VR headset was recommended since the plugin was built with Oculus' developer kit and only tested on an Oculus VR device.) During the study, participants can ask the researchers any questions about the plugin. After the participant submitted their executable game file and the documentation which illustrated the game design and the gameplay, we sent them a post-study questionnaire. The questionnaire contained System Usability Scale (SUS) questions [3] and several open-ended questions to ask about their overall experience regarding their developing experience with the SoundVizVR Plugin. The post-study questionnaire took approximately 15 minutes to complete. Finally, a follow-up contextual interview was conducted to clarify details (e.g., design intends, implementation methods) in their submitted games and discuss their experiences.

4 FINDINGS

We have the following findings according to the comments provided in the questionnaires, interviews, and through observation in the development process.

4.1 VR Game Development

In order to make the in-game sounds accessible, the participants could choose to either integrate the SoundVizVR Plugin into their existing VR projects or build a new VR game with accessible sound content utilizing the plugin's features.

D1 had a comprehensive experience in game prototyping and VR game development using the Unity platform. However, he had limited experience in accessible content creation for VR. Before our study, he indicated that he was interested in exploring the SoundVizVR Plugin's features, so he planned to build a VR game from scratch after receiving the plugin. D1 developed a VR treasure hunt game based on sound clues (Figure 2 [Left]). In this game, players were required to search for animals that were hidden/trapped in boxes. An instruction indicated the user to find an animal, and as clues, the animals occasionally made calls from the boxes. The boxes with different animals were randomly distributed in the scene. The players must move around in the scene with the VR controller's joystick and localize the required animal according to its calling sound. To make the audio accessible, D1 used the SoundVizVR Plugin with the icon representation method for the sound indicators above a sounding object and on the HUD Mini-Map. Thus, DHH players can localize the animal calls in the scene and identify the calling sound with the visualization method.

As a professional game developer, D2 had produced several VR games. However, he had limited experience making game audio accessible except for adding captioning for speech. In the study process, D2 brought a VR volleyball game he had made earlier and aimed to improve its audio accessibility via the SoundVizVR Plugin (Figure 2 [Right]). In the game, players must bounce the volleyball the opponent served in a designated basket using a VR controller. D2 visualized the interaction sounds of a volleyball bounce and a goal. In this case, DHH players can be notified of the location

and the existence of these interaction sounds when related events happen.

4.2 Ease of Use

Before the study, both developer participants mentioned they did not have accessibility experts in their development team. Thus, when searching for an accessibility tool for game developers, its ease of use is a crucial metric. In the post-study questionnaire, D1 and D2 rated 82.5 and 90 SUS scores respectively for the SoundVizVR Plugin, which indicated both of their adjective ratings [2] for the plugin had achieved "Excellent". In addition, D1 stated *"In terms of the current features of this plugin, it is straightforward to use. A user does not require much programming knowledge to use it."* Similarly, D2 reported that the plugin was easy to integrate into a game, especially compared to his previous experience working with an accessibility development tool for making subtitles.

Meanwhile, feedback from the participants indicated some designs of the plugin could be improved. D1 reported that while visualizing sounds' loudness, the sound indicators of the plugin did not consider the in-game volume setting of a sound source. For the current plugin's design, the size of the sound indicators was only bound with the average absolute value from the audio source's sample data. This design offered less flexibility when assigning different volume settings for the game objects that played the same sound clip asset. In addition, D1's VR game involved visualizing many sound source objects. D1 stated that the sound indicators binding took him some effort. "Suppose a developer needs to bind spatial sound sources with the SoundVizVR Manager individually through the Unity Inspector View. In that case, it will be cumbersome to use when involving many binding objects." Here, a more dynamically indicators assigning process of the SoundVizVR Manager would be ideal for the game designers and developers. "I recommend assigning a Unity Script Component⁵ to every game object and letting the SoundVizVR Manager collect their information automatically at runtime, rather than manually importing objects into the SoundVizVR Manager in this current design," D1 added.

4.3 Willingness of Making Game Content Accessible

In the pre-study questionnaire, both participants indicated that only a fraction of their past projects included accessibility features (D1: 20 - 40%; D2: < 20%). They reported that the only accessibility feature they implemented in past projects was captioning in role-playing and adventure games, aiming to make the in-game narrative text and dialog accessible for diverse player groups. In addition, D2 reported that when implementing game accessibility features, many of those features were hard to test because of the difficulties in recruiting target testers and lacking accessibility experts in the development team. Apart from that, D1 indicated he was very willing to make games accessible, while D2 leaned toward a neutral point of view. When asked about what prevented them from using accessibility tools during game development, both participants mentioned that finding a suitable tool for their games was challenging.

⁵Unity Scripting API: Component

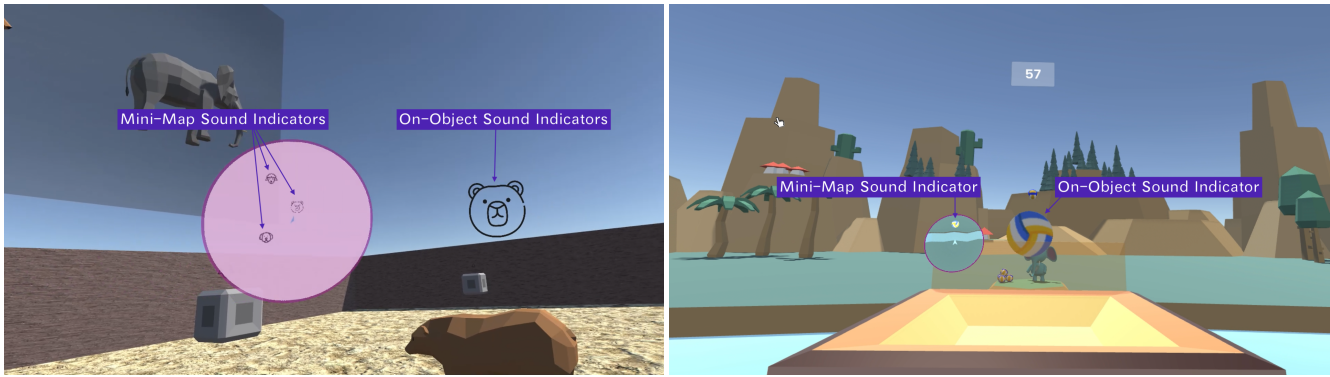


Figure 2: The in-game views of the games developed by D1 and D2. (Left) The VR treasure hunt game was developed by D1 with the SoundVizVR Plugin’s features enabled. (Right) The VR volleyball game was developed by D2 with the SoundVizVR Plugin’s features enabled.

After using the SoundVizVR Plugin in the study, both participants indicated that the development experience with the plugin had changed their attitudes toward accessibility. For example, D2 claimed, “Before using the plugin, I did not think much about accessibility when I designed and developed a game, even though there are some features in my game that enhance accessibility. For example, when players bounce the volleyball into the basket, I added a sound effect and a ‘+1’ floating icon. This ‘SoundVizVR Plugin’ can help to create and control the effects like this easily. After using the plugin, I feel the importance of accessibility, and I would definitely like to learn more and use some plugins like this to enhance accessibility.” A similar statement was supported by D1: “I think this plugin definitely can help the developers improve the accessibility easily.”

4.4 Introducing Accessibility Features in Development Workflow

One of our goals was to explore in which development phase the developers would consider introducing accessibility features to a game, as accessibility was often considered as an afterthought [31]. In the pre-study survey, D2, who had less than 20% of his past work included accessibility features, reported a low priority of implementing the features. (“I only consider accessibility features until the game is ready to release.”) D1 had 20% to 40% of his past work included accessibility features. He stated that his games would have different priorities based on their game genre and gameplay. However, both participants reported they planned to implement accessibility features when polishing the product because they considered the accessibility features, especially the captions, as additions to their indie games.

After using the SoundVizVR Plugin, D1 reported that he would plan to integrate this accessibility tool in the prototyping or polishing phases based on the game design. For example, for the sound-based treasure hunt game he produced, he introduced the tool in the prototyping stage because he wanted to evaluate the potential effect of using the plugin on the game balance. Thus, he wished to adjust the game design according to the offered features in the tool. On the other hand, D2 only visualized the interaction sounds in

his VR volleyball game, so he stated this plugin did not affect his development experience. However, he also planned to introduce the plugin in the prototyping stage as he realized the introduction of accessibility plugins does not affect the development process. Meanwhile, planning it in the prototyping phase meant he could use the accessibility features to make the game more engaging for DHH players.

4.5 Integrating Accessibility Features with Existing Content

Prior to the study, D1 stated that his primary concern in introducing accessibility features to a game was the effect on the game design and balance, while D2 wished the accessibility tool had a flexibility that could provide customization options to players for optimizing the inclusiveness.

In the post-study interview, D1 claimed the SoundVizVR Plugin fitted well with his sound-based VR treasure hunt game. However, since the critical design of the game involved sound localization and identification, players were required to guess which animal was making the sound they heard. Although he used the icon representation method for the sound-type indicators in the game as an implicit clue, he was still curious about if it would cause a significant difference in the game’s difficulty, either with the accessible feature enabled or disabled.

Besides, the concern from D2 was more about the HUD Mini-Map of the SoundVizVR Plugin. Since D2 integrated the accessibility tool into his volleyball game, he indicated that the Mini-Map did not fit well with the VR environment of the game. He stated, “It (the Mini-Map) may distract the user more or less. This may be because I designed a sports game that needs the players to respond to the game events quickly.” Here, he suggested providing more customization options for the map’s appearance or allowing developers to disable the Mini-Map base on the scenario. In addition, he wished the plugin could have more visual effects for the sound-type indicator, which might improve the engagement of the target players, especially when the indicator was hinting interaction sounds.

5 DISCUSSION

5.1 Optimizing Plugin Customizability

The SoundVizVR Plugin has provided options for customization, like the sound source objects that will be visualized, appearance configurations for the HUD Mini-Map and sound indicators, and the game objects captured by the Mini-Map (to determine what will be displayed on the map). However, the existing options only provide capabilities for developers to integrate the plugin into the game's user interface instead of fitting it into the gameplay. For example, in this study, a developer participant (D2) requested options of adding visual effects to the icons on the sound indicators, which aimed to make DHH players more engaged by enhancing the visual feedback of the interaction sound. Thus, a more customizable sound indicator or more programmable API provided by the SoundVizVR Manager prefab is preferred. In our practice, increasing the customizability and flexibility of the plugin could enhance the development experience while integrating it into the gameplay and keep players from being distracted by the accessibility components.

5.2 Challenges in Finding Accessibility Development Tools

Due to the lack of accessibility experts in the development team, indie game developers may seek professional development tools when implementing accessibility features in their games. However, developer participants reported that finding appropriate accessibility development tools was difficult. For example, in the Unity Asset Store⁶, where a Unity developer typically searches for game assets and development tools, there were limited accessibility plugins, not to mention those designed for VR. As identified in this preliminary study, the barrier of finding accessibility tools is one of the significant factors preventing indie game developers from making their game content accessible.

5.3 Planning Game Accessibility at An Early Stage

The developer participants had only implemented captions as an accessibility feature in their past projects. Since the captions had minimal impact on their gameplay design, prior to this study, the participants believed that accessibility features could be considered until the game polishing phase. However, the developer participants changed their minds after working on their games with the SoundVizVR Plugin. They agreed that the sound accessibility features should be planned during the game prototyping phase, and one of them thought he might also consider it during the development phase. The participants claimed that, planning accessibility features during the prototyping phase do not complicate the game development process. Moreover, planning accessibility early making it possible to identify potential conflicts between the accessibility features and the game design at an early stage so that the game designers can modify the game design accordingly.

5.4 Limitations and Future Work

Our preliminary study investigated integrating the SoundVizVR Plugin in the design process with a limited number of indie VR

game developers. Therefore, we still wish to explore it with developers of large commercial games, which may provide insights that cover more aspects, such as the game accessibility challenges and accessible VR game design practices in the game industry. Besides, our current work only focused on exploring the usability of the plugin from the VR developer's perspective. So, we suggest to conduct a user study among DHH players to evaluate the improvement of the VR game accessibility with the SoundVizVR Plugin integrated to provide feedback from players' perspective. In addition, the current design of the "SoundVizVR" system's Mini-Map sound indicators only focused on revealing spatial sound sources from a 2D top-down view, while visualizing 3D spatial information of the sounding objects with this system remained unexplored [13]. Therefore, we also aim to investigate and improve the SoundVizVR Plugin's design to enhance its capability to visualize spatial sound sources in 3D space [19, 26].

6 CONCLUSION

Prior research presented a sound visualization system called "SoundVizVR" to improve sound accessibility in VR projects [13]. Based on the proposed method and findings from that previous study, we developed a "SoundVizVR Plugin" for the Unity platform which enabled developers to use the SoundVizVR system in their VR applications and games. In this work, we conducted a week-long exploration with two game developers to evaluate the usability of the SoundVizVR Plugin. According to the observation and analysis of the study process, we provided game design practices for making sounds in VR games accessible and several suggestions on optimizing the developer experience of a sound accessibility plugin for the Unity platform.

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⁶Unity Asset Store

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